

Dataset Analysis for Decision-Making

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sicor_municipal_co... .csv
16.7 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

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To analyze the dataset and address any specific analytical questions, I will first load and examine the `sicor_municipal_contracts.csv` file to understand its structure, contents, and key characteristics. Once the dataset is understood, I can proceed to answer specific analytical questions using Python code.

Below is the step-by-step process:

Step 1: Load and Explore the Dataset

I will use Python's `pandas` library to load the dataset and inspect its structure. This includes checking:

- The number of rows and columns.
- Column names and data types.
- Missing values or anomalies.
- Summary statistics for numerical columns.

Here is the Python code to perform this exploration:

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sicor_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Display basic information about the dataset
8 print("Dataset Overview:")
9 print(data.info())
10
11 # Display the first few rows of the dataset
12 print("\nFirst 5 Rows:")
13 print(data.head())
14
15 # Check for missing values
16 print("\nMissing Values:")
17 print(data.isnull().sum())
18
19 # Summary statistics for numerical columns
20 print("\nSummary Statistics:")
21 print(data.describe())
```

Step 2: Key Characteristics of the Dataset

After running the above code, I will summarize the key characteristics of the dataset, such as:

- The total number of records (rows) and fields (columns).
- The types of data stored in each column (e.g., categorical, numerical, dates).
- Any patterns or trends observed in the data.
- Potential issues like missing values or outliers.

Step 3: Address Specific Analytical Questions



Once the dataset is understood, I can address specific analytical questions. Please provide a specific question you would like answered using this dataset. For example:

- What is the total value of contracts awarded in a specific year?
- Which municipality has the highest number of contracts?
- What is the average contract value by category?

For each question, I will write Python code to extract and process the relevant data and provide a clear, concise answer.

Example: Hypothetical Question

Question: What is the total value of contracts awarded in the year 2022?

Answer and Code:

Assuming the dataset contains a column named `contract_value` for the monetary value of contracts and a column named `award_date` for the date the contract was awarded, here is how we would calculate the total value of contracts awarded in 2022:

```
python
1 # Convert the 'award_date' column to datetime format
2 data['award_date'] = pd.to_datetime(data['award_date'], errors='coerce')
3
4 # Filter the data for contracts awarded in 2022
5 contracts_2022 = data[data['award_date'].dt.year == 2022]
6
7 # Calculate the total value of contracts awarded in 2022
8 total_value_2022 = contracts_2022['contract_value'].sum()
9
10 print(f"Total value of contracts awarded in 2022: ${total_value_2022:,.2f}")
```

Next Steps

Please provide a specific analytical question, and I will tailor the analysis and code accordingly. If there are any particular aspects of the dataset you want to focus on, let me know!

How does the state influence the total area allocated to investments?



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To address the question of how the state influences the total area allocated to investments, we need to analyze the dataset to determine whether there is a relationship between the state (or region) and the total area allocated for investment projects. This involves grouping the data by the state column (assuming such a column exists) and summing up the total area allocated for investments within each state.

Here's how we can approach this:

Step 1: Assumptions About the Dataset

We assume the dataset contains the following relevant columns:

- `state` : The state or region where the investment is located.
- `area_allocated` : The total area (e.g., in square meters or another unit) allocated to the investment.

If these columns are named differently, I will adjust the code accordingly after inspecting the dataset.

Step 2: Analysis Plan

1. **Group Data by State:** Aggregate the data by the `state` column.
2. **Sum the Area Allocated:** Calculate the total `area_allocated` for each state.
3. **Sort and Visualize Results:** Sort the states by total area allocated in descending order and optionally visualize the results using a bar chart.

Step 3: Python Code

Below is the Python code to perform this analysis:

```
python
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
```

```
4 # Load the dataset
5 file_path = 'sikor_municipal_contracts.csv'
6 data = pd.read_csv(file_path)
7
8 # Inspect the dataset to confirm column names
9 print(data.info())
10 print(data.head())
11
12 # Assuming the relevant columns are 'state' and 'area_allocated'
13 # Group by 'state' and calculate the total area allocated
14 state_area_summary = data.groupby('state')['area_allocated'].sum().reset_index()
15
16 # Sort the results in descending order of total area allocated
17 state_area_summary = state_area_summary.sort_values(by='area_allocated', ascending=False)
18
19 # Display the top states by total area allocated
20 print("\nTotal Area Allocated by State:")
21 print(state_area_summary)
22
23 # Optional: Visualize the results
24 plt.figure(figsize=(10, 6))
25 plt.barh(state_area_summary['state'], state_area_summary['area_allocated'], color='skyblue')
26 plt.xlabel('Total Area Allocated')
27 plt.ylabel('State')
28 plt.title('Total Area Allocated to Investments by State')
29 plt.gca().invert_yaxis() # Invert y-axis for better readability
30 plt.show()
```

Step 4: Explanation of Results

The output will provide:

1. A table showing the total area allocated to investments for each state, sorted in descending order.
2. A bar chart visualizing the same information for easier interpretation.

Step 5: Interpretation

From the results, we can observe:

- Which states have the highest and lowest total areas allocated to investments.
- Whether there are significant disparities in investment allocation across states.
- Potential patterns or trends that might suggest regional priorities or policies influencing investment distribution.

Missing Information or Ambiguities

If the dataset does not contain the assumed columns (`state` and `area_allocated`) or if additional clarification is needed (e.g., units of measurement for area), please provide more details about the dataset structure or the specific analytical focus.

